Dhruv Prajapati

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4-year **Mechanical Engineering** student with experience in **thermal engineering**, hardware validation, and failure analysis across energy, manufacturing, and battery technologies.

Education

Rochester Institute of technology — Rochester, NY
Bachelor of Science in Mechanical Engineering, Senior Year, Dean's List 2026, GPA 3.50 — ***Aug. 2023 – Dec 2027***

Technical Skills

Thermal & Fluid Systems: Heat Transfer, Fluid Mechanics, CFD (SolidWorks Flow Simulation, ANSYS Fluent)
CAD Design & Analysis: SolidWorks, ANSYS Mechanical & Fluent, CATIA v5
Programming: Python, MATLAB, Simulink, LabVIEW, Arduino
Hardware & Testing: Hitachi SEM, Oscilloscopes, Keysight Waveform generator, PIV, Instron
Manufacturing: CNC Milling, Lathe, GD&T, 3D Printing, Surface Metrology

Experience

Battery Cell Engineer Intern — **Aug. 2026 - Dec. 2026**
Tesla — *Austin, TX*

- Incoming intern working with 4680 tabless cell mechanical modeling & design, manufacturing and process development teams to support cell development.

Failure Analysis Engineer Intern — **Jun 2026 – Aug 2026**
Pacific Gas & Electric Company, PG&E - ATS — *San Ramon, CA*

- Numerical Analysis of Substation Post Insulators | *ANSYS Fluent + Maxwell + Structural, SolidWorks*
- Developed coupled CFD (transient), electromagnetic, and structural simulations using ANSYS to evaluate 4 operational loading conditions and identify critical stress concentrations in substation post-insulator assemblies.
- Supported validation of 80 in tall L-type 230 kV Lapp porcelain insulators through mechanical testing, data analysis, and technical reporting used for engineering assessment.

- Materials Characterization | *SEM, Microscopy, MATLAB, Root Cause Analysis*
- Conducted microscopy investigations on more than 5 failed utility assets, including switches, conductors, and post insulators, to identify fracture mechanisms and support root-cause analysis.

- Residual Moisture Drying model | *MATLAB, Experimental test*
- Developed a transient MATLAB evaporation model to predict residual moisture removal across a 14.7–1 psia pressure range, performing sensitivity studies on temperature and relative humidity effects after hydrostatic pipeline testing.

Mechanical Test Engineer Intern — **Aug. 2025 – Dec. 2025**
MKS Instruments — *Rochester, NY*

- Liquid-cooled Heat Sink Analysis | *SolidWorks Flow Simulation, experimental test bench*
- Designed and executed experimental characterization of a liquid-cooled microchannel heat sink for a 5 kW pulsed-DC generator (RPG-50), evaluating pressure-drop and flow-rate tradeoffs.
- Validated results using CFD simulations and first-principles analysis, contributing to a 25% improvement in system efficiency.

- Cross-functional projects | *ALT, Engineering Analysis, LabVIEW*
- Supported instrumentation and validation activities across multiple reliability test environments, including HVAC controls, liquid-nitrogen systems, and ALT/HASS/ESS chambers used for hardware qualification.

Research

Mechanical Engineer Researcher — **Feb. 2026 – May 2026**
Fluids Acoustics Structures Laboratory, FAST LAB — *Rochester, NY*

- Designed and built an experimental vortex-ring test platform integrating a linear actuator, Arduino-based control system, and Pololu motor controller to investigate confined-flow fluid dynamics.
- Conducted Particle Image Velocimetry (PIV) experiments to quantify vortex-induced flow structures and impingement heat-transfer behavior in confined geometries.